Credit Card Fraud Detection using Machine Learning

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Springboard

Key Takeaways

1. Fraud detection is a highly imbalanced machine learning problem, with only **0.58%** fraudulent transactions out of approximately **1.3 million** transactions.
2. Behavioral transaction patterns were more **predictive** than demographic customer information.
3. Tree-based ensemble models, especially **CatBoost**, achieved strong fraud detection performance while maintaining a very low false positive rate.

Business Problem

- financial losses
- real-time detection challenge
- class imbalance problem
- false negatives cost high
- **Objective**: Build a machine learning model to identify fraudulent transactions while minimizing missed fraud cases.
- **Dataset**:
  - ~1.3 million transactions
  - fraud rate ~0.58%

Dataset Overview

Data source

Kaggle: Credit card transaction dataset

Main variables:

transaction amount, merchant, category

Timestamp, latitude / longitude, gender

Job, fraud label

Challenge

Highly imbalanced dataset

Data Cleaning & Feature Engineering

Cleaning	Engineering
Data Cleaning • Removed missing / invalid values • Converted timestamps • Standardized merchant names • Removed leakage/high-risk fields	**Feature Engineering** • Time Features • Behavioral Features • Log Transformations

Exploratory Data Analysis (EDA)

Fraud rate by merchant category

- shopping_net
- misc_net
- grocery_pos

Transaction Amount Distribution

- Normal vs fraud
  - Fraud – higher transaction amount
  - Heavy tail

Time-based patterns

- Night transactions more suspicious

Top fraudulent merchants

- 693 merchants
- Target encoding merchants

Modeling Strategy - Why not accuracy?

PR-AUC is especially important for imbalanced fraud datasets.

Model	ROC-AUC	PR-AUC
Logistic Regression	0.9119	0.4382
Random Forest	0.9809	0.8556
XGBoost	**0.9972**	0.8390
LightGBM	0.9866	0.8574
CatBoost	0.9955	**0.9043**

CatBoost selected for deployment

- Handles categorical variables effectively
- Strong performance on imbalanced tabular data
- Better PR-AUC
- More stable generalization

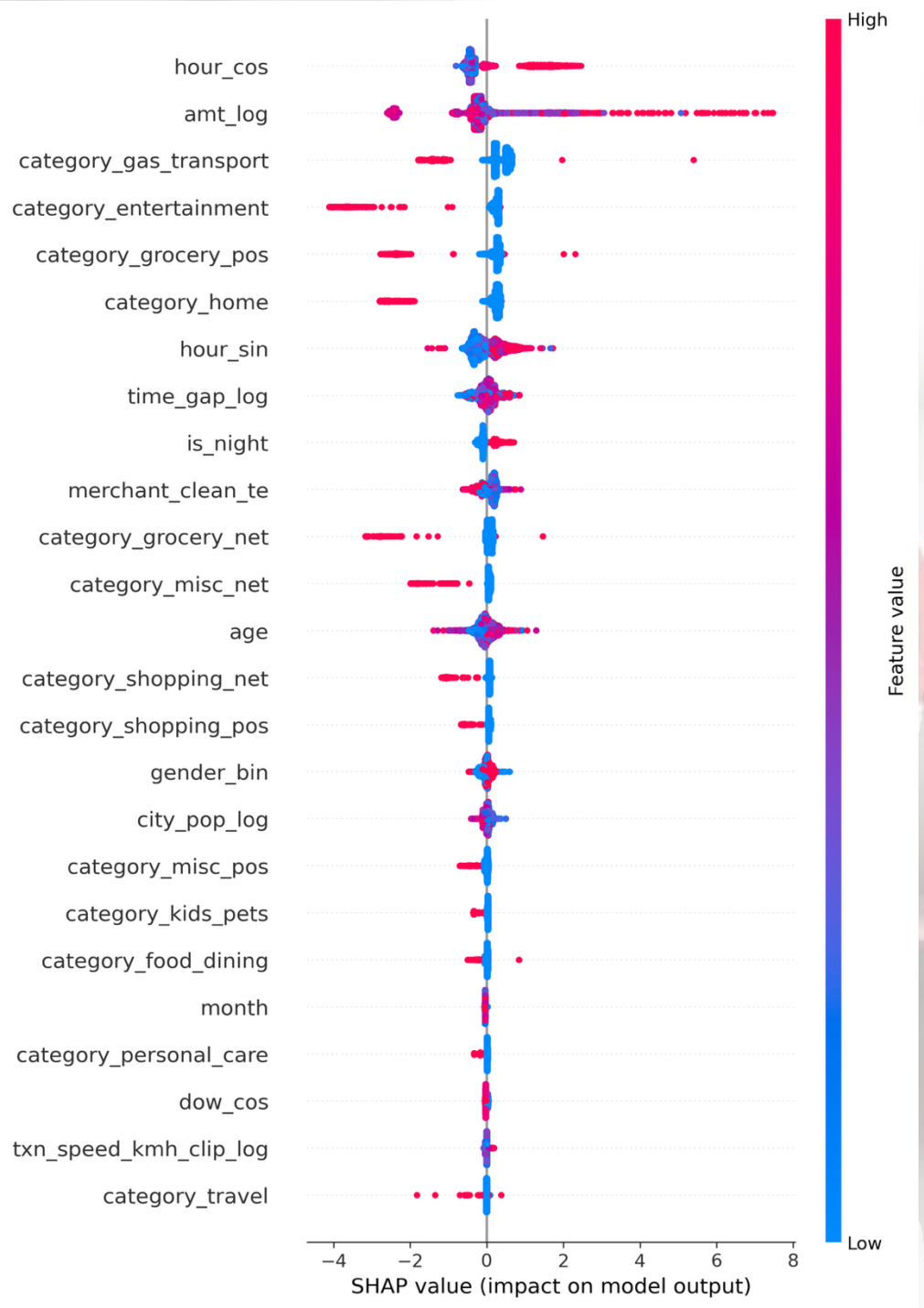
Final Test Deployment Results

Actual/Predicted	Legitimate	Fraud
Legitimate	553428	146
Fraud	543	1602

- Actual/Legitimate (regular transaction)
  - 553428 → True Negative (TN)
  - 146 → False Positive (FP)
- Actual/Fraud (fraudulent transaction)
  - 543 → False Negative (FN)
  - 1602 → True Positive (TP)
- False Positive Rate
  - $FPR = \frac{FP}{FP+TN}$
  - 146 / (146 + 553428) = 0.026%

Final CatBoost Test Performance

- ROC-AUC: 0.9940
- PR-AUC: 0.8762
- Precision: 0.92
- Recall: 0.75
- F1 Score: 0.8230



Feature Importance / SHAP

Top Behavioral Features

Transaction Amount
Transaction Speed
Merchant Category
Time Gap
Night Transactions

Behavioral features contributed more than demographic variables.

Limitations & Future Work

Limitations

- Historical dataset only
- No real-time streaming system
- Potential concept drift over time

Future Work

- Real-time deployment
- Graph neural networks
- Sequential/deep learning models
- Adaptive fraud thresholds

Conclusion

1. Feature engineering was critical.

2. Tree-based ensemble models outperformed baseline methods.

3. CatBoost achieved strong fraud detection performance on highly imbalanced data.

Thank You

- LinkedIn: https://www.linkedin.com/in/qinghao-c/
- GitHub: https://github.com/qchen077?tab=repositories